

DECK 19 · WEEK 10 · AFTERNOON 3

Alternative Investments

Private capital, real assets, hedge funds, and digital assets — what traditional long-only portfolios cannot do

CFA Level I · Vol 8 Learning Modules 1–7

Roadmap — Alternative Investments

7 sections · ~2 hours · Built bottom-up from CFA readings

01

Features, methods, structures

3 categories · fund / co-invest / direct · LP-GP term sheet & committed capital

02

Performance & fees

J-curve · IRR vs MOIC · net vs gross incentive fee · hard vs soft hurdle · HWM, clawback

03

Private capital — PE & private debt

VC / growth / buyout · LBO value bridge · exits · direct lending / mezz / distressed

04

Real estate & infrastructure

Core → opportunistic · cap rate vs REIT FFO multiple · BOT infra, greenfield / brownfield

05

Natural resources

Farmland & timberland · commodities via futures · contango vs backwardation

06

Hedge funds

L/S equity · event-driven · relative value · macro / CTA · 2-and-20, HWM, lockups

07

Digital assets

DLT, PoW vs PoS · crypto vs tokens · direct wallet / futures / ETFs · custody & regulation

Learning objectives

By the end of this deck you should be able to:

1. Classify any alternative investment into private capital, real assets, or hedge funds, and describe the access methods (fund / co-invest / direct) and their control-versus-fees trade-off.
2. Compute a GP's fees under 2-and-20 when the incentive fee is taken gross of the management fee, net of it, and net with a hurdle — and say which convention a term sheet is using.
3. Distinguish a hard hurdle from a soft hurdle, apply a catch-up clause, and explain what a high-water mark and a clawback each protect against.
4. Distinguish IRR from MOIC and explain why a high multiple over a long life can still be a poor return.
5. Categorise private equity (VC, growth, buyout) and private debt (direct lending, mezzanine, venture debt, distressed) and place each on the risk–return ladder.
6. Explain the shape of the commodity forward curve through cost of carry and convenience yield, and say which shape helps a long-only futures investor.

SECTION 01

Features, methods, structures

Alternatives are defined by what they are NOT — they lack the liquidity, transparency, and easy benchmarking of public equity and public debt

The curriculum lists THREE distinguishing features — and says those three features are what force the three structural consequences below.

Category	Sub-classes	What you are actually buying
Private capital	Private equity + private debt	Ownership of / lending to private firms
Real assets	Real estate, infrastructure, resources	Tangible: leases, tolls, crops, harvests
Hedge funds	L/S equity, event-driven, macro, RV	An approach, not an asset class

THREE FEATURES ⇒ THREE CONSEQUENCES

FEATURES (p. 6): 1. Specialised knowledge is needed to value the cash flows and the risks. 2. Typically low correlation of returns with traditional asset classes. 3. Illiquidity, long investment horizons, large capital outlays.

WHICH LEAD TO (p. 7): 4. Different investment structures, because direct investment is hard. 5. Incentive-based fees, to reduce the information asymmetry between manager and investor. 6. Performance-appraisal challenges.

Source: CFA Institute, Curriculum Vol 8, Learning Module 1, Alternative Investment Features, printed pp. 6–11.

Fund investing, co-investing, and direct investing trade control against effort — and the fee you pay falls at every step you take toward doing it yourself

The curriculum is explicit about the ladder: limited experience starts in funds, graduates to co-investing, and only the largest run direct.

FUND INVESTING

Commit capital to a GP. The FUND identifies, selects and makes the investments on your behalf.

- Fee: full management fee + full carry
- Control: lowest — no deal-level say
- For: investors without an in-house team
- Pre-commitment, then capital calls

CO-INVESTING

Invest in the fund AND hold co-investment rights to put more capital directly into the same deals.

- Fee: reduced management fee, often ZERO carry on the co-invest slice (Example 5)
- Control: deal-by-deal opt-in
- Cost: higher oversight burden on you

DIRECT INVESTING

Invest in the asset with no intermediary at all. Diligence, structuring and management are in-house.

- Fee: none — internal cost only
- Control: maximum
- For: the largest, most sophisticated — SWFs, big pensions

Source: CFA Institute, Curriculum Vol 8, Learning Module 1, Alternative Investment Methods, printed pp. 13–17; Example 5, p. 16.

A limited partnership concentrates capital from many LPs and all decision rights in one GP – the term sheet is where that imbalance is priced

LPs commit for 10 years; the GP draws the money down as deals close. The upfront cash outflow is a small fraction of the commitment.

ROLES AND LIABILITY

- GP = the fund manager
 - Sources, selects, manages and exits deals
 - Takes management fee + carried interest
 - Theoretically UNLIMITED liability
- LPs = outside investors
 - Passive; no role in day-to-day decisions
 - Liability capped at their investment
 - Accredited-investor tests apply
 - Capital drawn down over 3–5 years

TENDERLEDGE FUND VIII – EXHIBIT 3

- Maximum size: not to exceed USD 750m
- GP commitment: $\geq 2\%$ of aggregate LP commitments
- Initial closing: commitments $\geq$ USD 500m
- Final closing: 12 months after initial
- Term: 10 years from initial closing
- Management fee: 1.5% p.a. of each LP's COMMITMENT
- Hurdle: a HARD hurdle rate of 10%
- Performance fee: 20% of returns in excess of it
- Side letters: a most-favoured-nation clause — better terms granted to one LP go to ALL LPs

Source: CFA Institute, Curriculum Vol 8, Learning Module 1, Exhibit 3 (Tenderledge term sheet), printed pp. 15–16; LP/GP structure, pp. 19–21.

SECTION 02

Performance & fees

The private-capital J-curve is a fee artefact before it is a performance story — costs land in year one, value lands in year seven

Because cash flows are staged and at the GP's discretion, the curriculum treats IRR as the right starting point — the GP is being judged on timing as well as selection.

1 • COMMITMENT

Managers identify and select investments; capital is called immediately or with a delay.

Fees and expenses are incurred BEFORE deployment, and the assets generate little or no income.

Returns are usually negative here. This is the bottom of the "J".

2 • DEPLOYMENT

Funds are deployed: construction and improvements in real estate, turnaround spend in PE, operations for a start-up.

Cash outflows still typically exceed inflows, and management fees keep reducing returns.

Marks stay near cost — the accounting "middle period".

3 • DISTRIBUTION

The turnaround, build-out or start-up phase completes. Assets appreciate or generate income above cost.

Returns accelerate; the fund realises gains by exiting — trade sale, IPO, SPAC.

Returns then level off as capital is returned and the fund winds down.

Source: CFA Institute, Curriculum Vol 8, Learning Module 2, Exhibit 1 (Investment Life Cycle), printed p. 37; J-curve, p. 38.

IRR prices the timing of cash flows and MOIC ignores it — which is precisely why the GP quotes one and a careful LP asks for the other

MOIC treats one euro returned in year 2 exactly like one euro returned in year 15. Market practice: $DPI = \text{distributions} / \text{paid-in}$; $RVPI = \text{NAV} / \text{paid-in}$; $TVPI = DPI + RVPI$.

$$\text{MOIC} = (\text{Realised value} + \text{Unrealised value}) / \text{Invested capital}$$

CFA LM 2 KNOWLEDGE CHECK — HIMITSU INVESTS IN ZZZ

Year 0: invest JPY 3.8bn. Year 2: a further JPY 1.2bn. Year 3: a further JPY 0.2bn. Year 8: sell the stake for JPY 8.5bn.

1. Total invested capital = $3.8 + 1.2 + 0.2 = \text{JPY } 5.2\text{bn}$ (paid-in, less fees and expenses)
2. $\text{MOIC} = 8.5 / 5.2 = 1.63x$ ← this is the number the CFA question actually asks for
3. IRR: solve $-3.8 - 1.2/(1+r)^2 - 0.2/(1+r)^3 + 8.5/(1+r)^8 = 0 \Rightarrow r \approx 6.8\%$
4. Sanity check: $1.63^{(1/8)} - 1 = 6.3\%$; the later draws lift it only to 6.8%. So 1.63x sounds respectable and is a single-digit annual return.

MOIC = 1.63x · IRR ≈ 6.8%. A 2.0x over 15 years is only 4.7% p.a. — the multiple hides the clock.

¹ Curriculum defines invested capital as paid-in capital LESS management fees and fund expenses — so MOIC's denominator is net of fees, not gross commitments.

² The curriculum's summary table prints "IRR 20%" beside MOIC 1.63x. Solving NPV = 0 on the cash flows as printed gives 6.8%; 1.63x over 8 years cannot be 20%.

Source: CFA Institute, Curriculum Vol 8, Learning Module 2, MOIC and Equation 1, printed p. 39; Knowledge Check (Himitsu / ZZZ), p. 39. DPI, RVPI and TVPI are market practice, not Vol 8 content.

The incentive fee depends on a convention nobody states out loud – the same 2-and-20 on the same 30% year produces three different bills

Base first: hedge funds and REITs charge the management fee on AUM, private equity on COMMITTED capital. Then the convention: which terms sit inside that bracket.

$$R_{GP} = P_1 \cdot r_m + \max[0, (P_1 - P_0 - \text{fee} - \text{hurdle}) \times p]$$

CFA LM 2 KNOWLEDGE CHECK – AWJ CAPITAL, USD 100m, 2-AND-20, +30% GROSS

$P_0 = \text{USD } 100\text{m}$, $P_1 = \text{USD } 130\text{m}$. Management fee 2% on YEAR-END AUM. Incentive fee 20%. Same fund, same year, three fee conventions.

1. Management fee, identical in all three cases: $130 \times 2\% = \text{USD } 2.60\text{m}$
2. GROSS of the management fee: $(130 - 100) \times 20\% = 6.00\text{m} \Rightarrow \text{fees } 8.60\text{m}$, investor +21.40%
3. NET of the management fee: $(130 - 100 - 2.60) \times 20\% = 5.48\text{m} \Rightarrow \text{fees } 8.08\text{m}$, investor +21.92%
4. NET, plus a 5% hard hurdle: $(130 - 100 - 5 - 2.60) \times 20\% = 4.48\text{m} \Rightarrow \text{fees } 7.08\text{m}$, investor +22.92%

Fees 8.60 / 8.08 / 7.08 – same manager, same year. The convention is worth 152 bps.

A hard hurdle pays carry on the excess only; a soft hurdle pays carry on the whole return once the hurdle clears – and a full catch-up turns one into the other

Ignoring management fees throughout. USD 100m of capital, an 8% preferred return per annum, 80/20 carry, property sold at the end of Year 2 for USD 160m.

$$\begin{aligned} \text{HARD, no catch-up: } r_{GP} &= \max[0, p(r - r_h)] \\ \text{WITH catch-up: } r_{GP} &= \max[0, r_{cu} + p(r - r_h - r_{cu})] \end{aligned}$$

where: r = period return; r_h = hurdle; p = carry rate; r_{cu} = the catch-up, paid 100% to the GP until the cumulative split reaches 80/20.

Waterfall step	Soft – LP	Soft – GP	Hard – LP	Hard – GP
1 · Return of LP capital	\$100m	–	\$100m	–
2 · 8% preferred return, two years	\$16m	–	\$16m	–
3 · GP catch-up to a 20% share	–	\$4m	–	–
4 · Residual split 80 / 20	\$32m	\$8m	\$35.2m	\$8.8m
TOTAL PAYOUT	\$148m	\$12m	\$151.2m	\$8.8m

WHAT THE CATCH-UP ACTUALLY DOES

Full catch-up $\Rightarrow$ the GP takes 20% of the entire \$60m gain. Hard hurdle, no catch-up $\Rightarrow$ 20% of the \$44m excess only.

Source: CFA Institute, Curriculum Vol 8, Learning Module 1, hurdle definitions and Equations 1–2, printed pp. 23–24; Learning Module 2, Question Set 1–2, printed pp. 53–54.

High-water marks, clawbacks and the waterfall all answer one question — how do you stop the GP being paid twice for the same gain

A high-water mark protects against paying twice on a round trip. A clawback protects against paying early on winners that later losers wipe out. They are not the same tool.

HIGH - WATER MARK

The fund's peak value at a calculation date, NET of fees. Below it, no incentive fee until the peak is regained.

AWJ, same fund:

- Yr 1: \$100m → \$130m, HWM set at \$121.4m
- Yr 2: falls to \$110m — mgmt fee \$2.2m only, NO incentive fee
- Yr 3: \$128m — fee on $(\$128 - \$121.4) \times 20\% = \$1.32\text{m}$, not on the whole \$18m rebound

CLAWBACK

Gives LPs the right to RECLAIM performance fees already paid. Triggered when early exits win and later deals lose.

Tenderledge, \$20m in two deals:

- Arguston \$10m sold at \$22m ⇒ accrues $20\% \times \$12\text{m} = \2.4m
- Heartfield \$10m fails, aggregate gain falls to \$2m
- Fee due is $20\% \times \$2\text{m} = \0.4m ; \$2.0m is returned

THE WATERFALL

Deal-by-deal (AMERICAN): carry is taken per deal, so the GP is paid before LPs have their capital and hurdle back on the whole fund. GP-friendly.

Whole-of-fund (EUROPEAN): every distribution goes to LPs until capital and hurdle are met at fund level. LP-friendly.

In Exhibits 10 and 11 the fund breaks even overall: American pays \$13m then claws it all back; European pays nothing at all.

Source: CFA Institute, Curriculum Vol 8, Learning Module 1, printed pp. 24–26 (Exhibits 10–11); Learning Module 2, Example 4 (AWJ) p. 51 and Example 6 (Tenderledge clawback) pp. 49–50.

SECTION 03

Private capital – PE & private debt

Private equity is indexed to where a company sits in its life cycle – VC underwrites product-market fit, buyout underwrites operations

Venture capital IS a form of private equity — a specialised one, and a small share of the market by value. The exam tests that sentence.

Life-cycle stage	Revenues	Cash flow	Risk	Typical investor
Seed	None	Negative	Very high	Angel / pre-seed VC
Start-up	Low to none	Negative	High	Early-stage VC
Growth	Increasing	Increasing	High	Later-stage VC
Maturity	Positive, predictable	Positive, predictable	Low	LBO / buyout
Decline	Declining	Declining	Increasing	Buyout / distressed

WHAT SITS AT EACH END

EARLY: pre-seed (USD 5k–500k, mostly individuals) → seed (USD 25k–5m, the first stage VC funds usually enter) → early-stage → later-stage (USD 5m+, institutions). Convertible preferred is the standard instrument, because it ranks ahead of common in a liquidation.

LATE: growth / minority equity takes a non-controlling stake in a maturing firm; buyout takes control. Average hold is about five years, with a tail from under six months to more than ten.

Source: CFA Institute, Curriculum Vol 8, Learning Module 3, Exhibit 1 (life cycle), printed p. 70; Exhibit 2 (VC stages), p. 70; exits, p. 73.

A buyout creates equity value through exactly three levers — earnings growth, multiple expansion and debt paydown — and the bridge has to reconcile to the cent

Figures are illustrative. Real deals also carry transaction fees, add-on M&A, and a management incentive plan that dilutes the sponsor at exit.

TEXTBOOK BUYOUT — 5-YEAR HOLD, 60% DEBT, 6.0x ENTRY → 7.0x EXIT

EBITDA₀ = 100 at 6.0x ⇒ EV 600, funded 360 debt + 240 equity. EBITDA grows 6% p.a. Debt amortises to 210. Exit at 7.0x.

1. EBITDA₅ = $100 \times 1.06^5 = 133.8$ · Exit EV = $7.0 \times 133.8 = 936.8$
2. Exit equity = EV – net debt = $936.8 - 210.0 = 726.8$ ⇒ MOIC = $726.8 / 240.0 = 3.03x$
3. IRR = $3.03^{(1/5)} - 1 = 24.8\%$ p.a.
4. BRIDGE — earnings growth: $\Delta\text{EBITDA} \times \text{entry multiple} = 33.8 \times 6.0 = 202.9$
5. BRIDGE — multiple expansion: $\Delta\text{multiple} \times \text{exit EBITDA} = 1.0 \times 133.8 = 133.8$ · debt paydown = $360 - 210 = 150.0$
6. Check: $202.9 + 133.8 + 150.0 = 486.7 = 726.8 - 240.0$. ✓ The bridge closes.

MOIC 3.03x · IRR 24.8%. Of the 486.7 created, 27% is multiple expansion the sponsor did not earn.

Source: Illustrative worked example built for this session. The CFA curriculum (Vol 8, LM 3, printed pp. 68–69) describes LBO value creation qualitatively but gives no numerical build.

Exit route is chosen at entry, not at exit — and each route trades price against speed, certainty and disclosure

The curriculum treats trade sale and public listing as the two main routes; recapitalisation, secondary sale and write-off are grouped as "other".

Exit route	Key advantage	Key drawback
Trade sale	Synergy premium; fast, simple execution	Few buyers; regulatory scrutiny
IPO	Highest potential price; mgmt approval	Costly, slow, volatile, lockup
SPAC	Fixed valuation; forward guidance OK	Dilution; stockholder overhang
Direct listing	No underwriters, lower cost	Least common; raises no capital
Secondary sale (PE-to-PE)	Liquidity without public disclosure	Price set by other sponsors only
Recapitalisation	Extracts cash, keeps control; lifts IRR	Adds leverage; not a true exit
Write-off / liquidation	Frees the team; closes a failed deal	Marks a loss on the track record

THE TWO THAT ARE NOT REALLY EXITS

A recapitalisation adds leverage and pays the sponsor a dividend — the curriculum is explicit that it is NOT a true exit, because control is retained. A secondary sale only moves the asset from one sponsor to another.

Source: CFA Institute, Curriculum Vol 8, Learning Module 3, Exhibit 3 (pros and cons), printed p. 76; other exit strategies, p. 77.

Private debt has four major categories plus unitranche and specialty loans – and seniority, collateral and covenants matter far more than the coupon

Coupons float off a reference rate, so the headline yield moves with policy. Modelling private debt is hard: thin data and artificially smooth marks.

Category	Rank in the structure	Typical borrower	Defining feature
Direct lending	Senior, secured	Mid-market corporates	Fixed payment schedule
Mezzanine	Sub to senior, sr to eq.	LBOs, recaps, M&A	Warrants / conversion rights
Venture debt	Ranks above equity	Start-ups, early stage	Line of credit; equity rights
Distressed debt	Varies by tranche	Firms at/near default	Priced to expected recovery
Unitranche	Between senior and sub	Mid-market LBOs	One blended interest rate

THE ONLY THREE SPREADS THE CURRICULUM ACTUALLY PRINTS

Generic private debt quoted as SOFR + 375 bps (p. 82). · Peterburgh's syndicated commercial mortgages at MRR + 150 bps, first lien, 0.75 LTV, 15 years (p. 81). · Hertz debtor-in-possession financing at MRR + 725 bps on USD 1.65bn while in Chapter 11 (p. 82). Note the spread ladder that implies: senior secured real estate 150, unsecured corporate 375, lending into a bankruptcy 725.

¹ Curriculum: private lending funds filled the financing gap left by traditional lenders after the 2008 crisis; the excess return is an illiquidity premium.

² Vol 8 gives no spread table. Every spread on this slide is a figure the curriculum itself prints in an example, and is labelled as such.

Source: CFA Institute, Curriculum Vol 8, Learning Module 3, Private Debt Investment Characteristics, printed pp. 79–83; Exhibit 4, p. 80.

Risk and return rise as you climb the private capital structure — each rung names the specific premium you are being paid to accept

Private capital is not uncorrelated: Prequin indices correlate 0.63 to 0.83 with public indexes (Exhibit 7). The curriculum calls the benefit "moderate", not absent.

Rung — lowest risk first	What it is	The premium you are paid for
1 · Infrastructure debt	Most senior real-asset debt	Illiquidity; long duration
2 · Senior real-estate debt	First liens; IG CMBS	Illiquidity; collateral work
3 · Senior direct lending	Senior secured corporate loans	Credit risk of private firms
4 · Unitranche debt	Blended senior / subordinated	Credit + subordination
5 · Mezzanine debt	Junior to senior, above equity	Subordination + equity kicker
6 · Private equity, co-invest	Common equity in private firms	Leverage, illiquidity, concn.

VINTAGE YEAR IS THE OTHER AXIS

A fund's vintage is the year it makes its first investment; a typical fund runs 10–12 years, split into a roughly five-year investment period and a harvesting period. Funds seeded in a low-valuation recovery ride the recovery; funds that deploy into a high-valuation peak do not. The curriculum's remedy is not manager selection — it is diversifying ACROSS vintage years.

Source: CFA Institute, Curriculum Vol 8, Learning Module 3, Exhibit 6 (risk and return levels), printed p. 85; Exhibit 7 (correlations), p. 86.

SECTION 04

Real estate & infrastructure

Real estate runs a single spectrum from bond-like senior debt to equity-like opportunistic – and the further up you go, the more return comes from growth, not rent

The exhibit also splits the ladder a second way: the lower rungs are available through public vehicles, the upper rungs are primarily private.

Strategy	What sits in it (Exhibit 4)	Return character
Senior debt	First mortgages; investment-grade CMBS	Bond-like
Core	Stable income; long leases; sale-leaseback	Bond-like, lease-driven
Core-plus	Minimal refurbishment; tenant repositioning	Income + modest growth
Value-add	Vacant space; redevelopment; sub-IG CMBS	Cash-flow growth
Opportunistic	Development; mezz debt; unrated CMBS; distress	Price appreciation

THE FUND WRAPPER FOLLOWS THE STRATEGY

Core sits in INFINITE-LIFE, OPEN-END funds and REITs — investors contribute and redeem throughout, like a mutual fund, because the underlying cash flows are stable enough to support it. Opportunistic sits in FINITE-LIFE, CLOSED-END funds, because development and repositioning cannot be unwound to meet a redemption. Mismatching the two is how open-end property funds end up gating.

Source: CFA Institute, Curriculum Vol 8, Learning Module 4, Exhibit 4 (Real Estate Risk–Return Spectrum), printed p. 103; open-end vs closed-end funds, p. 101.

Direct property is valued off a cap rate on stabilised NOI and a REIT trades on an FFO multiple — the same building, two clocks, and they do not tick together

Curriculum point that anchors this slide: listed REITs are repriced continuously, private property is appraised perhaps once a year — the mismatch makes measured correlations artificially low.

DIRECT · CAP RATE ON NOI

Office block: gross rents USD 8m, operating expenses USD 2m, vacancy already reflected.

1. $\text{NOI} = 8 - 2 = \text{USD } 6\text{m}$ (no interest, no tax, no capex)
2. Comparable transactions imply a 5.5% cap rate
3. $\text{Value} = \text{NOI} / \text{cap rate} = 6 / 0.055$
4. = USD 109m. Cap rate up 50 bps to 6.0% $\Rightarrow$ USD 100m.

Value = USD 109m · a 50 bp move costs 8%

PUBLIC · FFO MULTIPLE

REIT: net income USD 1.60/sh, real-estate depreciation USD 0.80/sh, no property sold this year.

1. $\text{FFO/sh} = 1.60 + 0.80 = \text{USD } 2.40$
2. Depreciation is added back: buildings are not consumed
3. Peer group trades at 15x FFO
4. = $15 \times 2.40 = \text{USD } 36/\text{sh}$

Implied price = USD 36/sh

THE HONEST BRIDGE: mark the REIT's buildings at market cap rates, deduct the debt — that is NAV per share. Price versus NAV is the whole argument.

¹ NOI excludes financing costs, depreciation, income tax and capital expenditure — it is a pre-financing, pre-tax property-level number.

² Direct capitalisation (value = NOI / cap rate) assumes a stabilised, perpetual NOI. Use a DCF when the income is not yet stabilised.

Source: INSTRUCTOR EXTENSION, not Level I. Vol 8 (LM 4, printed p. 100) mentions FFO only as adjusting for depreciation, distributions and preferred dividends; it gives no formula, no cap rate and no NOI.

A REIT escapes double taxation by distributing 90-100% of taxable rental income — and the price of that exemption is that it cannot retain capital to grow

Three forms, one exemption. Curriculum's stated disadvantage of the listed wrapper: higher correlation with public equity than private real estate.

EQUITY REITs

Own properties outright or through partnerships and joint ventures.

- Strategy: maximise occupancy and rents, minimise operating and maintenance cost
- Distributes 90–100% of taxable net rental income
- Reports GAAP/IFRS EPS but is judged on FFO
- Return: dividend plus appreciation

MORTGAGE REITs

Underwrite loans against real estate or hold mortgage-backed securities.

- Owns the DEBT, not the building
- Return is a net interest margin, usually levered
- Sensitive to rates and to credit spreads
- Can be private or publicly traded

HYBRID REITs

Hold both equity and debt exposure to real estate in one vehicle.

- Return mixes rental and interest income
- Lower sector-specific risk than either pure form
- Less common than the two pure types

Why the listed wrapper at all: transparency, no direct property management, and no forced selling to meet redemptions.

Source: CFA Institute, Curriculum Vol 8, Learning Module 4, REIT forms and structure, printed pp. 99–101; Exhibit 3, p. 100.

Infrastructure is distinguished from real estate by public purpose, a concession contract, and cash flows that are contracted rather than negotiated

Core infrastructure cash flows look bond-like — regulated tariffs, availability payments, long-dated tolls — which is exactly what a liability-driven pension wants.

Category	Examples (Exhibit 6)	Cash-flow mechanism
Economic — Transportation	Roads, bridges, tunnels, airports, ports	Usage / traffic – market risk
Economic — ICT	Telecom towers, data centres	Leases to carriers
Economic — Utility, energy	Grid, power, water, gas, LNG, waste	Tariffs or take-or-pay
Social — Education, health	Schools, hospitals, prisons	Availability payments
Social — Housing, buildings	Social housing, municipal buildings	Availability payments

THE CONTRACT IS THE ASSET

A public–private partnership uses a special purpose entity to raise debt and equity, build and operate the asset, and hand it back or sell it at the end of a concession agreement. The distinction that matters for risk: DEMAND-based revenue (a toll road, an airport) carries traffic risk you cannot hedge, while AVAILABILITY-based revenue pays you for keeping the asset to standard whether or not anybody uses it. Exhibit 8 puts availability-based social infrastructure at the low-risk end and demand-based greenfield at the high-risk end.

Source: CFA Institute, Curriculum Vol 8, Learning Module 4, Exhibit 6 (Categorizing Infrastructure Investments), printed p. 108; concession and PPP, pp. 107–109.

Infrastructure has its own J-curve — greenfield builds through negative cash flows, brownfield buys the operating history, and secondary-stage buys only the cash flow

The curriculum ranks the risk explicitly: secondary-stage lowest, brownfield incrementally riskier, greenfield riskiest of the three.

GREENFIELD · RISKIEST

Develop a new asset that does not yet exist. Treated by the curriculum as an opportunistic investment.

BOT life cycle:

- BUILD — long approval and construction phase, cash flows negative
- OPERATE — concession agreement sets the revenue parameters
- TRANSFER — handed to government, sold on, or decommissioned

Usually done alongside a developer or strategic partner.

BROWNFIELD · MIDDLE

Expand or upgrade an EXISTING facility. Often arrives via privatisation, or as a sale-leaseback of a completed greenfield project.

- Some financial and operating history exists
- Shorter investment period, immediate cash flows
- Still needs capital expenditure
- Attracts both strategic operators and financial investors seeking stable long-term returns

SECONDARY-STAGE

Fully operational asset requiring no further material investment or development over the holding period.

- Immediate cash flow, steady and predictable
- Bond-like track record already established
- Lowest expected return of the three
- Natural home for pensions, insurers and SWFs buying duration

Some assets never get here — they keep needing capital.

Source: CFA Institute, Curriculum Vol 8, Learning Module 4, Exhibit 7 (Build-Operate-Transfer model), printed p. 110; stages of development, pp. 109–110.

SECTION 05

Natural resources

Raw land, farmland and timberland share real-estate DNA but take their value from the soil and the biology, not from anything built on top

The defining difference from real estate: there is little or no focus on physical improvements. Soil quality, climate and geology set the value instead.

Attribute	Raw land	Farmland	Timberland
Return drivers	Price of land	Harvest, crop prices, land	Growth, lumber prices, land
Direct revenue	Appreciation; leases	Crop sales; leases	Timber sales; leases
What sets value	Physical location	Location, cycle, soil	Location, quality, phase
Main risks	Best alternative use	Weather, climate, disease	Weather, climate, disease
Typical owner	Mostly institutional	Mostly individuals	Mostly institutional
Ownership forms	Direct / partnership	Direct / partnership / REIT	Direct, REIT, TIMO

TIMBERLAND IS BOTH THE FACTORY AND THE WAREHOUSE

The curriculum's own phrase (p. 126). Timber is GROWN — the factory — and can be STORED simply by not harvesting it — the warehouse. Farm products must be harvested when ripe, with little flexibility. That asymmetry is a real option on the lumber price, and it is the reason timberland behaves differently from every other agricultural asset. About 98% of US farmland is family owned; institutions hold roughly a quarter of the global investable timberland base.

Source: CFA Institute, Curriculum Vol 8, Learning Module 5, Exhibit 1 (Raw Land, Farmland and Timberland), printed p. 125; land vs real estate, pp. 124–125.

Commodity exposure is bought through futures, so the SHAPE of the forward curve — not just the spot price — decides what a long-only investor earns

Commodities themselves pay no cash flow. You hold a contract, and every month that contract has to be rolled into the next one.

$$F_o(T) = S_o \cdot e^{((r + c - i) \times T)}$$

where: r = risk-free rate; c = cost of carry — storage, insurance, transport; i = convenience yield, the non-cash benefit of holding the physical, which falls as inventories rise.

BACKWARDATION · CURVE DOWN

Spot ABOVE forward. Convenience yield exceeds cost of carry, which happens when inventories are LOW.

Rolling forward means selling the expiring contract dear and buying the next one cheap $\Rightarrow$ roll return POSITIVE.

CONTANGO · CURVE UP

Spot BELOW forward. Cost of carry exceeds convenience yield, which happens when inventories are HIGH.

Rolling forward means selling cheap and buying dear $\Rightarrow$ roll return NEGATIVE, month after month.

TOTAL RETURN = SPOT RETURN + ROLL RETURN + COLLATERAL RETURN

Spot = the price move. Roll = the gain or bleed from rolling down or up the curve. Collateral = interest on the cash backing the futures, so a futures programme earns the policy rate on its margin. Curriculum Example 1: in April 2020 NYMEX crude futures went below zero — demand collapsed, producers could not cut fast enough, storage filled, and a curve in backwardation in January (Exhibit 4) was in deep contango by May (Exhibit 5).

¹ The three-way decomposition in the orange band is standard market practice and CFA Level II / CAIA material. Vol 8 gives Equation 1 and the rule of thumb only.

² Curriculum rule of thumb, verbatim: contango generally LOWERS the return of a long-only investor; backwardation ENHANCES it.

Source: CFA Institute, Curriculum Vol 8, Learning Module 5, Equation 1 and the backwardation/contango rule, printed pp. 132–133; Example 1 (crude oil, 2020), p. 133.

Commodities are the only asset class here whose returns swing hard on the inflation regime — farmland and timberland diversify but do not hedge inflation

Inflation is split at the median US CPI of 2.26%. The gap for commodities between the two regimes is 32 percentage points; for every other asset class it is a few.

Metric	Global stocks	Global bonds	Commodities	Farmland	Timberland
Annualised return, 30 years	6.89%	4.39%	7.81%	10.95%	8.69%
Annualised standard deviation	16.76%	6.14%	24.39%	5.88%	6.76%
Return, HIGH-inflation years	+9.40%	+5.66%	+22.87%	+12.78%	+10.44%
Return, LOW-inflation years	+5.43%	+4.18%	−9.26%	+9.85%	+5.70%
Correlation with global stocks	1.00	0.15	0.41	0.12	0.02
CORRELATION WITH INFLATION	0.10	−0.22	0.54	−0.17	0.02

READ THE BOTTOM ROW BEFORE THE TOP ONE

Only commodities correlate meaningfully with inflation (0.54). Farmland at −0.17 and timberland at 0.02 do not — the curriculum says so explicitly. What farmland and timberland offer is DIVERSIFICATION: 0.12 and 0.02 against global stocks, and roughly zero against bonds. And note why commodities score as a hedge: not because they do brilliantly in high inflation, but because they do badly in low inflation. That is an expensive insurance premium in the years you do not need it.

¹ Windows differ by exhibit: returns and volatility are Q3 1992–Q2 2022 quarterly; inflation splits are 1993–2021 annual; correlations are 1990–Q1 2020 quarterly.

² Commodities are S&P GSCI Total Return, heavily energy-weighted; farmland and timberland are appraisal-based NCREIF indexes, so their volatility is understated.

Source: CFA Institute, Curriculum Vol 8, Learning Module 5, Exhibit 6 (p. 136), Exhibit 7 (p. 137), Exhibit 8 (p. 138) and Exhibit 10 (p. 140).

SECTION 06

Hedge funds

Hedge funds are classified by investment approach, not by what they own — which is why the strategy taxonomy, not the asset class, is the right map

The curriculum's own count: FIVE broad categories. Not every hedge fund shorts or uses leverage — many simply exploit a niche expertise.

Strategy family	Sub-strategies in Exhibit 1	Defining characteristic
Equity hedge	Long/short equity, market neutral, short biased	Bottom-up; variable beta
Event-driven	Merger arb, distressed, special sits, activist	Corporate catalyst; long bias
Relative value	Convertible arb, fixed-income arb, multi-strat	Levered spreads, neutral
Opportunistic	Global macro, managed futures (CTAs)	Top-down; directional
Multi-manager	Fund of funds, multi-strategy	Diversified across managers

WHAT ACTUALLY DISTINGUISHES A HEDGE FUND (LM6, PRINTED P. 157)

Seven characteristics, and the curriculum numbers them: 1. lighter legal and regulatory constraints; 2. flexible mandates permitting shorting and derivatives; 3. a larger investment universe; 4. aggressive styles allowing concentrated positions in credit, volatility and liquidity risk premiums; 5. relatively liberal use of leverage; 6. liquidity constraints — lockups and gates; 7. relatively high fees combining management and incentive components.

Source: CFA Institute, Curriculum Vol 8, Learning Module 6, five broad categories and Exhibit 1 (Hedge Fund Strategies), printed p. 151.

Equity long-short is the original hedge fund strategy — and over the curriculum's own ten-year sample the market index beat every single variant of it

Beta is chosen deliberately, from long-biased fundamental growth down through market neutral to short bias. The correlations show it worked as designed.

Strategy	Avg m/m	Std dev	Corr. vs idx	Where the alpha is meant to come from
Market index (S&P 500)	+0.30%	2.0%	1.00	The bar every strategy must clear
Fundamental value	+0.03%	1.0%	0.80	Turnarounds; short the overpriced
Fundamental growth	+0.03%	1.5%	0.67	High growth vs failing models
Market neutral	-0.04%	0.5%	0.22	Pure selection; needs leverage
Short biased	-0.50%	2.3%	-0.72	Forensics: accounting flaws

WHAT THE TABLE IS ACTUALLY SAYING

The correlations behave exactly as the strategy labels promise: value and growth are long-biased at 0.80 and 0.67, market neutral sits at 0.22, short bias at -0.72. But the return column is the uncomfortable one — the index averaged +0.30% a month and no strategy came close. Market neutral delivers the low volatility it advertises (0.5% monthly) and needs leverage to turn that into a return worth paying for. Short-biased managers have struggled to post meaningful long-term returns across 30 years of generally rising markets.

Source: CFA Institute, Curriculum Vol 8, Learning Module 6, Example 1 (Tenderledge Investments — Equity Strategies), monthly data over 10 years, printed p. 153.

Event-driven strategies trade a corporate catalyst and relative value trades a temporary mispricing — both earn thin spreads, so both use leverage

Curriculum detail worth keeping: event-driven strategies tend to be LONG-biased, with merger arbitrage the least biased of them. A merger takes 6–24 months; a bankruptcy can take years.

EVENT-DRIVEN

- Merger arbitrage — buy the target below the announced takeover price, short the acquirer. Profit as the deal spread narrows to close. The spread exists because of legal, regulatory and walk-away risk; shorting the acquirer also expresses overpayment risk. Primary risk: the deal breaks.
- Distressed / restructuring — buy discounted debt still senior enough to be covered by assets, or buy the FULCRUM SECURITY that converts to equity in the reorganisation.
- Special situations — spin-offs, asset sales, rescue finance, buybacks.
- Activist — take a stake big enough for a board seat, then push for divestitures, distributions or a change of management. Operates in PUBLIC equity, which is what separates it from private equity.

RELATIVE VALUE

- Convertible bond arbitrage — buy the convertible, short the delta-equivalent amount of the issuer's stock. Conceptually market-neutral; exposed to bankruptcy risk, which can be hedged with puts or CDS.
- Fixed income, general — sovereign relative-value rates and investment-grade credit. Trades between issuers, between corporate and government, between parts of one issuer's capital structure or curve.
- Fixed income, asset-backed / mortgage-backed / high yield — hunting relative security and quality mispricing in higher-yielding paper.
- Multi-strategy — trade relative value across asset classes, redeploying capital between pods as conditions change.

Source: CFA Institute, Curriculum Vol 8, Learning Module 6, event-driven and relative value strategies, printed pp. 154–156.

Hedge fund terms do three jobs at once — set the fee, set the legal wrapper, and set how long your money is trapped when you want it back

Curriculum principle: redemption terms should match the liquidity of the assets. Where they do not, a routine drawdown turns into a forced sale.

F E E S

Standard "two and twenty":

- 2% management fee on AUM — not on committed capital
- 20% of net profits to the GP
- High-water mark: no incentive fee until the previous peak is regained

Emerging "1 or 30":

- The GREATER of a 1% management fee or 30% of alpha versus a benchmark

Fund of hedge funds adds a second layer, typically 1 and 10, on top of the underlying fees.

S T R U C T U R E S

- MASTER-FEEDER — an onshore feeder and an offshore feeder both invest into one master fund. Built for tax efficiency across investor types.

- SEPARATELY MANAGED ACCOUNT — the investor creates the vehicle and the assets are held and registered in the investor's own name; day-to-day management is delegated.

- FUND OF ONE — a hedge fund created for a single investor.

Side letters address one investor's legal, tax or reporting needs and can supersede the fund documents.

R E D E M P T I O N T E R M S

- LOCKUP — the minimum holding period before any withdrawal is permitted.

- SOFT LOCKUP — early redemption against a fee payable to THE FUND, not the manager, so remaining investors are compensated.

- NOTICE PERIOD — typically 30 to 90 days, so the manager can liquidate in an orderly way.

- GATE — limits or restricts redemptions for a period, usually at the manager's discretion.

Redemptions cluster when performance is poor. That is the point.

Source: CFA Institute, Curriculum Vol 8, Learning Module 6, fee structures and forms, printed pp. 160–162; liquidity provisions, Learning Module 2, printed p. 44.

SECTION 07

Digital assets

Distributed ledger technology replaces a trusted intermediary with a consensus protocol — and the protocol you choose is a trade of energy against openness

A blockchain is a set of rules for deciding which block joins the chain and becomes the immutable truth. Everything else follows from that one design choice.

PROOF OF WORK (PoW)

The next block is decided by a computationally costly lottery. Miners spend real energy solving a cryptographic puzzle and are paid in the network's own asset.

- Security comes from the ECONOMIC COST of attacking
- Participants accept the LONGEST chain as truth
- To rewrite history you must out-compute the whole network — the 51% attack threshold
- Even then, only a limited length of history can be altered
- Energy-intensive; this has been the most widely used protocol

PROOF OF STAKE (PoS)

Selected participants — validators — PLEDGE CAPITAL to vouch for a block's validity. A majority of other stakers must then attest to it.

- Security comes from the economic cost of LOSING THE STAKE
- Validators are paid both for proposing and for attesting
- Far lower energy consumption than PoW
- Same majority-control boundary applies: the protection is that no malicious party gains a majority of the stake

Under both protocols, successful validation is rewarded with newly issued digital assets — this is mining.

Source: CFA Institute, Curriculum Vol 8, Learning Module 7, Proof of Work and Proof of Stake, printed pp. 183–184; Exhibit 3 (permissioned vs permissionless), p. 185.

Digital assets split into cryptocurrencies and tokens — and the investment case, the regulator and the failure mode are different for each row

Bitcoin and Ether together were over 80% of total cryptocurrency market value as of July 2022. Everything below them is a long tail.

Family	Type	What it is for
Cryptocurrencies	Bitcoin	Medium of exchange, store of value
Cryptocurrencies	Altcoin – Ether, Solana	Programmable chain, smart contracts
Cryptocurrencies	Altcoin – stablecoins (USDC)	Peg to fiat – a settlement rail
Cryptocurrencies	Altcoin – memecoins	Community hype; no stated use case
Cryptocurrencies	Central bank digital currency	Tokenised fiat, central-bank issued
Tokens	Security token	Digitised ownership of a security
Tokens	Utility token	Pays for services on the network
Tokens	Governance token	Votes on protocol changes
Tokens	Non-fungible token (NFT)	Unique authenticated digital object

HOW YOU BUY IT — Direct: a wallet on a centralised exchange (liquid, but a security target) or a decentralised one (hard to attack, hard to regulate). Indirect: cash-settled crypto futures, ETFs holding futures rather than coin, listed miners and exchanges, and crypto hedge funds.

Source: CFA Institute, Curriculum Vol 8, Learning Module 7, Exhibit 4 (Different Digital Assets), printed p. 185; investible digital assets, pp. 190–192.

Digital-asset risk is not a list of drawdowns — it is a list of ways the position goes to zero for reasons unrelated to the price

The curriculum is blunt: cryptocurrency trading venues are largely unregulated, and both centralised and decentralised exchanges raise investor-protection concerns.

RISK 1 · CUSTODY

- A centralised exchange holds your private keys. If its servers are compromised, trading halts and the custodial wallet keys leak.
- Self-custody moves the risk to you: a lost seed phrase is an unrecoverable loss, with no issuer and no registrar to appeal to.
- A decentralised exchange is far harder to attack — and far harder to regulate or to sue.

RISK 2 · REGULATION

- Venues are largely unregulated, so market manipulation is possible and common — the curriculum's own pump-and-dump example is EthereumMax.
- Whether a token is a security or a commodity is unresolved, and it decides the supervisor.
- Jurisdictional patchwork: legal in some countries, banned outright in others.
- The curriculum asks for clear, predictable standards ACROSS jurisdictions. They do not yet exist.

RISK 3 · VALUATION

- No cash flows and no consensus on how to value them. The curriculum links that missing anchor directly to the volatility.
- Algorithmic pegs can fail fast: TerraUSD was backed only by Luna. In one week Luna supply went from 350 million to 6.5 trillion coins; the peg broke on 9 May 2022.
- Collateralised stablecoins are not exempt — Terra's collapse triggered a run on Tether and USDC over reserve transparency.

Source: CFA Institute, Curriculum Vol 8, Learning Module 7, exchanges and fraud, printed pp. 195–196; Terra/Luna and stablecoin reserves, pp. 192–193; Exhibit 7 (FTT), p. 199.

Deck 19 — Alternative Investments

- Alternatives are defined negatively. Three features — specialised knowledge, low correlation, illiquidity — force three consequences: different structures, incentive fees, and appraisal difficulty.

- A hard hurdle pays carry on the excess; a soft hurdle pays on the whole return once cleared. A full catch-up makes a hard hurdle economically identical to a soft one — GP \$12m vs \$8.8m on the same \$60m gain.

- IRR prices timing and MOIC ignores it: 1.63x over eight years is 6.8% a year. Ask for both, plus DPI, before believing any track record.

- Real assets: the fund wrapper must match the asset — open-end for core, closed-end for opportunistic. Cap rate is a yield, so a 50 bp move costs 8% of value; a REIT reprices that in a morning, an appraisal takes a year.

- Fee arithmetic turns on the BASE and the CONVENTION. Hedge funds charge on AUM, private equity on committed capital; an incentive fee taken gross of the management fee, net of it, or net with a hurdle gave 8.60 / 8.08 / 7.08 on the same 30% year.

- High-water marks stop double payment on a round trip; clawbacks reclaim carry paid on early winners; American waterfalls pay per deal, European wait for the whole fund.

- LBO equity value = earnings growth + multiple expansion + debt paydown, and the bridge must reconcile. In the worked deal, 27% of the value created was multiple expansion the sponsor did not earn.

- Commodity futures returns depend on the CURVE: backwardation adds roll return, contango bleeds it. Commodities hedge inflation (0.54); farmland and timberland diversify but do not hedge it.